

Cybersecurity

Commanding the Command Line

Lesson 3.3



Learning Objectives

By the end of today's class, you will be able to:

- 1 Explain how options modify the default behavior of a terminal command.
- 2 Use the **man** command to list instructions and options for each command.
- 3 Use the **find** command to locate files based on search parameters.
- 4 Use the **grep** command to search within the contents of files.
- 5 Combine multiple commands in sequence with pipes for intermediate IT tasks.



Welcome



Activity:

Warm-Up

Continue in your role as a security analyst at Candy Corp.

- Candy Corp's management apprehended Ruth and Henry, and they're grateful for all your work so far.
- The local authorities have video evidence that Sugar Corp delivered a cash payment to Candy Corp's back door on October 13, 2019.
- You are tasked with gathering physical access logs to prove Henry or Ruth opened the back door for the delivery.

Suggested Time:

15 Minutes

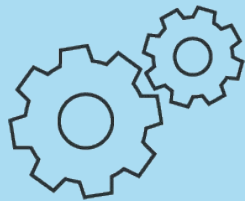


Time's up!
Let's review



Questions?





The **find** Command

The find Command

Looking for our files

- Using `cd` to navigate the file structure is time consuming.
- As a cybersecurity professional, you often are not given the location to:
 - User files
 - Logs
 - Configuration files



The `find` Command

The `find` command searches for files and directories with one command.



Will

By default, `find` will search through the current directory and the subdirectories within that current directory.



Will not

However, `find` does not review the contents within a file; it only reviews the file name or directory name.



find Syntax

This command finds **all** files.

```
find -type f
```



The option **-type** and required parameter **f** indicate that we are searching for files.



find Syntax

This example finds a **specific** file.

```
find -type f -name log.txt
```

The option `-name` searches for an exact match of the specified parameter `log.txt`.



find Syntax

Finding a file with case insensitivity, change the `-name` option to `-iname`.

```
find -type f -iname log.txt
```

This will find the files called `log.txt` (lowercase) or `LOG.TXT` (uppercase).



find Syntax

The symbol known as a **wildcard** searches for all files ending with `.txt`.

The `*` wildcard symbol indicates that any file ending with `.txt` will be displayed regardless of what comes before `.txt`.

```
find -type f -iname `*.txt`
```

With the addition of `-iname`, this example finds all files that end with `.txt` or `.TXT` in your current directory and its subdirectories.



find Syntax

Search for a file located in another directory.

Specifically, we're looking for the case-insensitive `log.txt` in the `/root/desktop` directory.

```
find /root/desktop -type f -iname `log.txt`
```

Place the desired directory after the `find` command and before the `-type` option.



Syntax for Finding a Directory

Add **d** as the parameter for the option **-type**.

```
find -type d
```

```
find -type d -name logs
```

```
find -type d -iname logs
```

```
find -type d -iname `*1013`
```

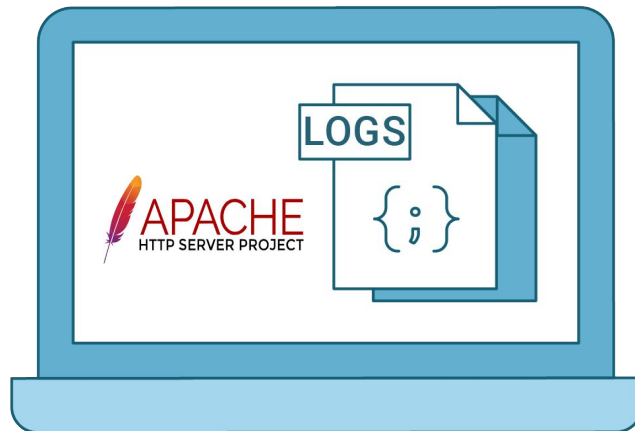
```
find /root/desktop -type d -iname logs
```

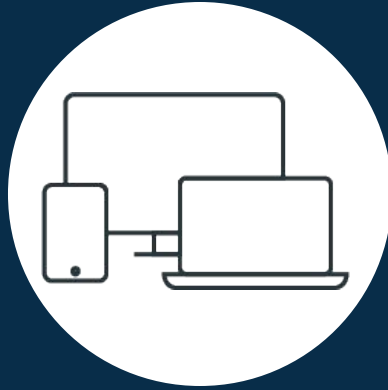


Demonstration Scenario

Your manager at ACME Corp has tasked you with finding logs for a certain type of web server called **Apache** for the date of October 13.

- The directory should be named **apache** and the log files should have the date noted as **1013** in their file names.
- Since there are many directories, you will use the **find** command to complete these tasks.





Instructor **Demonstration**

find

Demo Summary

`find` simplifies the searching process

<code>find</code>	Command-line command used to locate a file or directory
<code>type f</code>	An option used to find files
<code>type d</code>	An option used to find directories
<code>name</code>	Additional option used to find specific file or directory names
<code>iname</code>	Additional option used to find case-insensitive names



Activity:

find Your Way

Continue in your role as a security analyst at Candy Corp.

- Your manager has identified why PeanutButtery.net was targeted: Henry had hidden a secret recipe for a revolutionary candy within the file system, and Sugar Corp was trying to find it.
- Your manager at Candy Corp needs you to search through the files and file directories of the PeanutButtery.net server to find the hidden recipes in its file system.

Suggested Time:
20 Minutes



Time's up!
Let's review

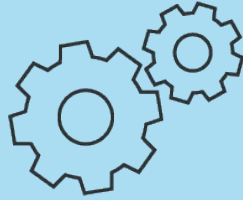


Questions?





Break
15 mins



The grep Command



The **find** command only searches for the names of files, **not** the contents.

grep

global **r**egular **e**xpression **p**oint is used to search for data inside files.

grep by default:

Returns the **entire line** that
the desired data is found

Only searches for data in the
current directory





grep Syntax

In this example, `grep` finds the lines in which user bob is mentioned in the file `log1.txt`.

```
<grep data_point File_to_search_inside>
```



grep Syntax

Use `grep` to find a specific data point within a single file.

Command

The file being searched for the data point

```
grep bob log1.txt
```

The specific data point searched for

This command displays all the lines where the data point **bob** was found inside the file **log1.txt**. If no matches of **bob** are found in the file, nothing is returned.



grep Syntax

Use `grep` to find a specific data point within **multiple** files.

Find where `bob` exists in all `.txt` files.

- `bob` is the specific data point being searched for.
- `*` indicates that it will search through all files that end with `.txt`.

```
grep bob *.txt
```

The command will display all the `.txt` files where the value of `bob` was found, followed by the lines where it was found.



What does this
command do?

```
grep -i bob *.txt
```



grep Syntax

Find a **case-insensitive** specific data point.

This command finds the line where the user **bob** or **BOB** exists within all **.txt** files.

```
grep -i bob *.txt
```



grep Syntax

Find a **filename** containing a specific data point.

This command outputs the file names of `.txt` files containing `bob` or `BOB`.

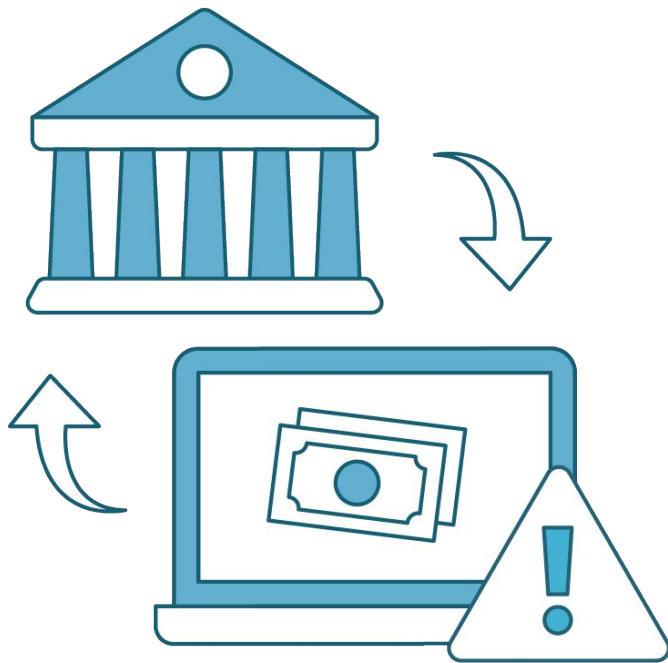
```
grep -i1 bob *.txt
```

Note: `-i1` are two separate options.

- `i` indicates case insensitivity.
- `1` indicates to only return the file name.
- `i` and `1` use a single hyphen.

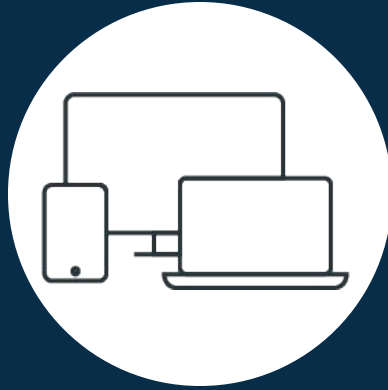


Demonstration Scenario



Your manager asks for your help with a security investigation into an illegal money transfer that took place on May 17.

- They tell you the suspect, Sally Stealer, stated that she never logged in to the company's banking website and definitely did not transfer any money on May 17.
- You must use the `grep` command to search the application logs to see if Sally Stealer logged in on that day and, if so, whether she transferred any funds.



Instructor **Demonstration**

grep

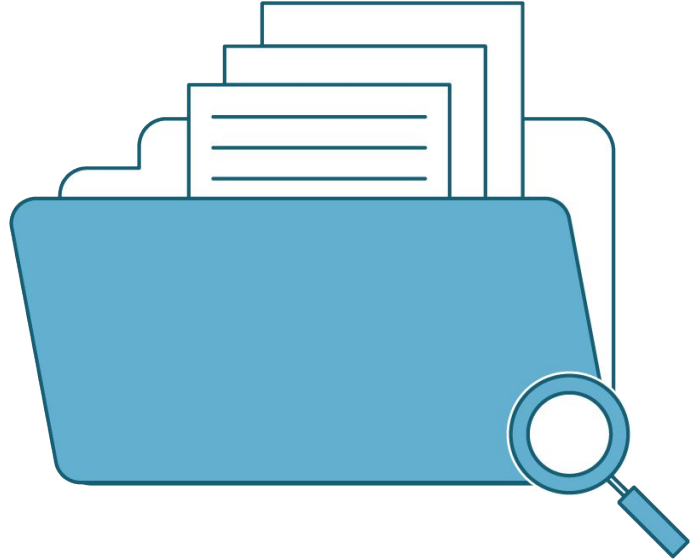
Demo Summary

`grep` is used to find data points within a file

Basic syntax:

```
grep data_point file_to_search
```

- `i` indicates case insensitivity.
- `l` indicates to only return the file name.





Activity:

grep

Continue in your role as a security analyst at Candy Corp.

- Your manager found out that Sugar Corp recently made a large purchase of guavaberries. Sugar Corp currently does not have any candy that includes guavaberries, but Candy Corp does.
- Your manager suspects that Sugar Corp is trying to reproduce some of Candy Corp's secret recipes that use guavaberries.
- Your task is to determine which of the secret recipes contain guavaberries in their list of ingredients.

Suggested Time:
20 Minutes

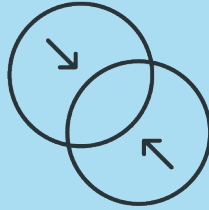


Time's up!
Let's review



Questions?





Combine Commands with Piping

Terminal Review

Commands we've discussed for working with files:

find

Searches for **file** names
or **directories**



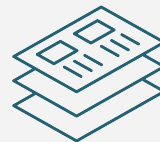
grep

Searches for **data**
points inside files



wc

Counts lines or words
inside a file



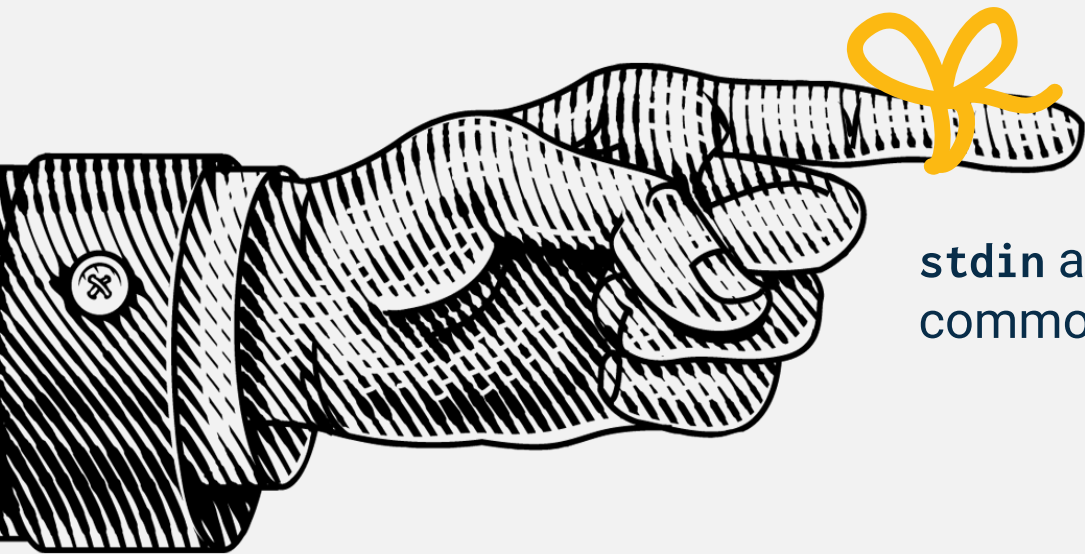


Security professionals
are often tasked with
combining commands
to complete certain
tasks.

Pipes

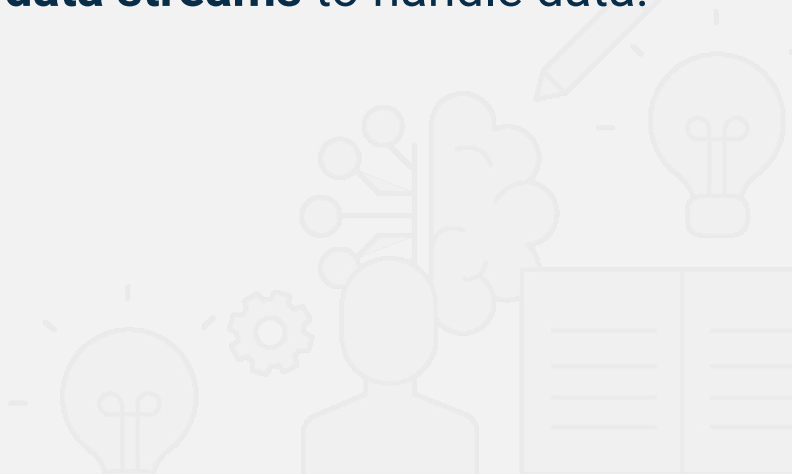
Use pipes to combine commands in a single line.

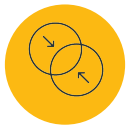
-  Redirects the output of commands to another to complete additional tasks
-  Designated by the symbol: |
-  Multiple can be used in a single command
-  Process data flows from left to right



Remember,

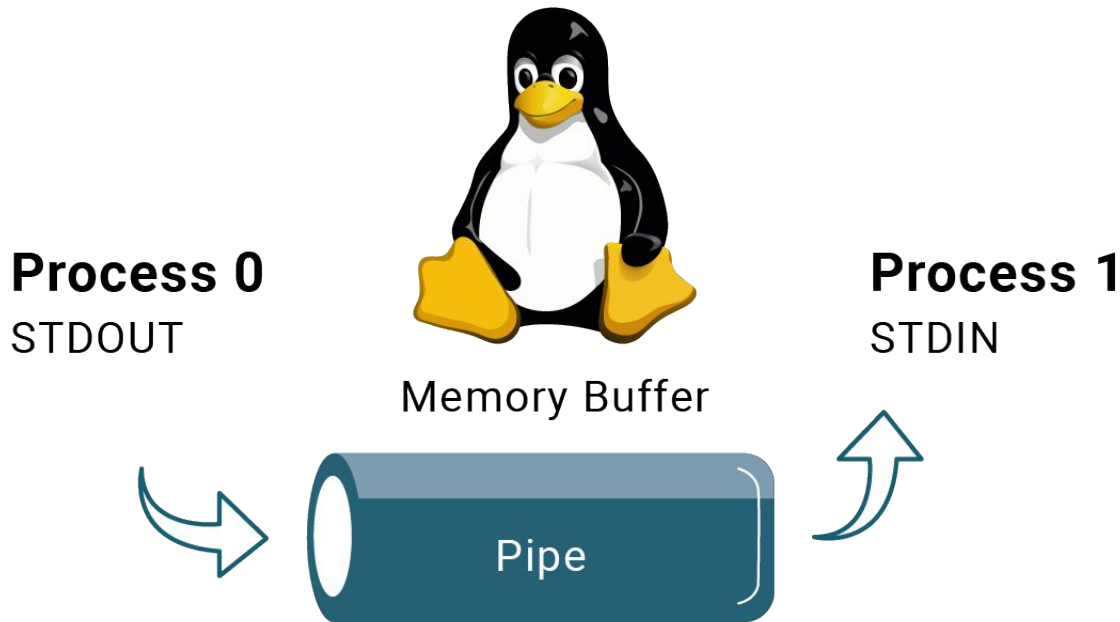
`stdin` and `stdout` are our most common **data streams** to handle data.





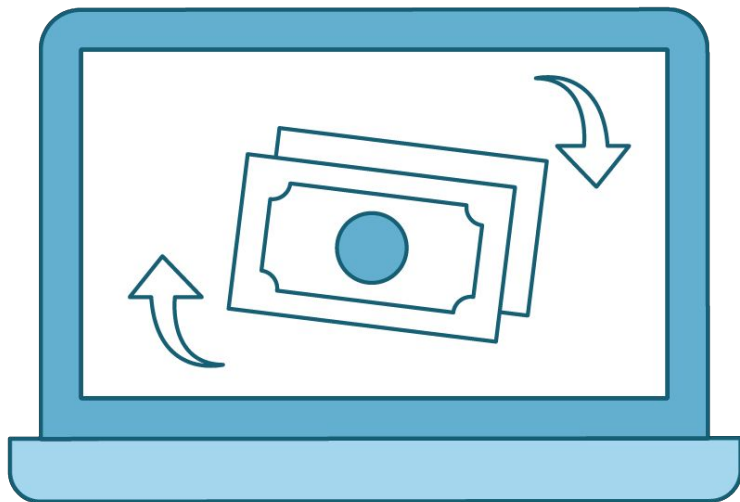
Combine Commands with Piping

Pipes are used with data streams



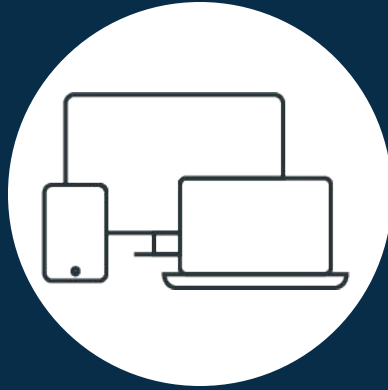


Demonstration Scenario



Your manager has tasked you with continuing the investigation against Sally Stealer. They believe she may have transferred other large amounts of money.

- Your manager created a single file called **largetransfers.txt** that contains all transfers over one million dollars.
- Your manager asks you to count how many of those transfers belong to Sally Stealer.



Instructor **Demonstration**

Pipes

Demo Summary

Pipes redirect the output of a command to another

01

A pipe is designated with the | symbol.

02

Multiple pipes can be used in a single command.

03

Pipes are unidirectional, meaning the processing of the data flows from left to right through the pipeline.



Activity:

Gather Evidence

Continue in your role as a security analyst at Candy Corp.

- Candy Corp believes it has enough evidence to send to the authorities to charge Sugar Corp with a cyber crime.
- Your task is to gather several points of evidence from your file systems that can be provided to the authorities to prove that Sugar Corp is stealing data.

Suggested Time:
20 Minutes



Time's up!
Let's review



Questions?





Next class

We will learn about permissions, file ownership, and access controls.



Thank you